



Spun Fair, Done Elsewhere: A Household Chore Wheel's Assigned Turn Against Who Completed It, Before and After an Appeals Process for Disputed Turns

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Abstract. Household chore wheels assign a rotating turn to a family member for a set of recurring tasks, and families that keep one routinely treat its posted name as an accurate record of who is responsible for a given week's work. Whether the wheel's assignment matches who actually performs the task is rarely checked directly. We report an 18-week field deployment across 76 households already using a chore wheel, independently logging four tracked chores each week against the wheel's posted assignee before and after a formal appeals process was introduced for disputed turns. Assignment concordance averaged 61.2% across the nine-week baseline and rose to 87.5% across the nine post-appeals weeks, most of the increase arriving within the first two post-appeals weeks. Task-completion rate, whether a chore was done at all by anyone, held flat across the same boundary (91.0% against 90.5%, not significant). Appeals filed under the new process concentrated heavily on a single chore (68.1% named bin duty), and requiring a second adult's counter-signature to approve an appeal left post-appeals completion statistically unchanged from single-adult approval. We read the results as consistent with the appeals process improving the accuracy of the record more than the underlying division of labour, and propose the appeals log itself, not the concordance rate it produces, as the more informative object for a household reviewing its own wheel.

Introduction

A household chore wheel assigns a rotating turn — dishes, bins, floors, pet care — to a named family member, typically for a week at a time, and most households that keep one treat its posted name as sufficient record of who is responsible for that week's task. Unlike a formal roster in a workplace or classroom, the wheel is rarely audited against what actually happened: no one logs whether the named assignee performed the chore, only whether the wheel was spun. What a family's informal division-of-labour instrument does or does not track, once it is treated as authoritative, has been studied at length for gendered and parental housework more broadly [[1]; [2]; [3]], but the wheel itself — a device many households already run — has not, to our knowledge, been validated as a measurement instrument in its own right.

A further complication arises when a family adds a formal appeals process to its wheel: a mechanism, familiar from workplace grievance and voice procedures [[4]; [5]], by which a disputed turn can be contested and reassigned. Once an appeal succeeds, the wheel's posted name changes to match whoever will actually perform the task, which mechanically raises any measure of concordance between the wheel and the doer — whether or not the underlying division of labour has itself become fairer. This paper reports a field validation designed to separate those two possibilities.

This paper's contributions, each stated with the caution it deserves:

- We provide, to our knowledge, the first field validation of a household chore wheel as a measurement instrument, quantifying the gap

between the wheel's posted assignment and independently logged completion under naturalistic conditions.

- We show that introducing a formal appeals process substantially closes this concordance gap, though our design cannot on its own establish whether the closure reflects a fairer division of labour or a more accurate paper trail.
- We find that the household's task-completion rate, independent of who performs it, does not measurably change alongside the concordance improvement, and that appeals concentrate heavily on a single chore, a pattern our results suggest is more consistent with preference-avoidance than with a general fairness correction.

Related work

Two literatures bear on the wheel rather than the chore. The first concerns how household labour is measured and allocated. [3] reviews the methodological challenges of measuring routine family work, cautioning that different measurement approaches to the same household can yield materially different pictures of who does what. [1] and [2] trace how housework time is distributed across household members and how that distribution has shifted, and [6] compares mothers' and fathers' time with children specifically, a parallel division-of-labour question one step removed from chores proper. Children's own participation in household labour has a comparable literature: [7] weighs socialization against household need as competing explanations for what children are assigned, [8] find that children track and remember perceived differential treatment between siblings closely, and [9]'s review situates that sensitivity within sibling relationships more broadly. [10] documents parents' attitudes to the allowance ledger, an informal household instrument that, like the chore wheel, converts a task into a recorded transaction. [11]'s recent systematic review confirms the broader measurement question remains unsettled a decade after [3] first raised it.

The second concerns what a formal appeals mechanism corrects, and how visibly. [5] find that giving a party voice in a decision raises perceived fairness substantially independent of the decision's actual outcome, a finding [4]'s later construct-validation work situates within

a wider, multidimensional account of organizational justice that includes procedural, not just distributive, fairness. Applied to a household wheel, where the appeal's outcome is directly observable by every party rather than mediated by an institution, that independence is exactly what our design tests for. [12]'s review of the Hawthorne effect cautions that the fact of being studied can change what a record shows independent of any change in the underlying behaviour, while [13]'s review of common method bias cautions separately that a self-report record can diverge from the behaviour it purports to measure for reasons unrelated to observation at all. Self-monitoring instruments are, in most domains where they have been studied, positively associated with the behaviour they track ([14]), which makes our result — a record's accuracy climbing with no detectable behavioural change beneath it — a departure worth flagging rather than a predictable outcome. A prior Slop University audit of a shared office fridge's QR-label disposal app found label compliance climbing sharply against an unmoving dispute rate [15], a divergence this paper's design was built to test for directly in a domestic setting; a companion audit of an apartment lobby noticeboard found entries classified as a settled ruling closing markedly faster despite no resident holding any enforcement authority over one [16], consistent with a record acquiring legitimacy independent of what it actually resolves.

Method

Setting and participants. Households were recruited through three primary-school parent networks and a local family-services newsletter on the criterion that they already maintained an active, wall- or fridge-mounted chore wheel in weekly use, with at least two children old enough to be assigned a turn and two resident adults able to alternate verification duty. Seventy-six households met these criteria and completed the full 18-week deployment.

Baseline concordance (weeks 1–9). Each week, whichever adult set the wheel photographed its assignment for one of four tracked chores (dishes, bins, floors, pet care) at the Sunday reset. The household's other resident adult, blind to the photographed assignment, independently logged who actually performed each chore by the following Saturday close. A turn was scored

concordant when the wheel's named assignee matched the independently logged doer:

$$C_w = \frac{1}{N_w} \sum_{i=1}^{N_w} \mathbb{1}[a_i = d_i],$$

where a_i is the wheel's assignee for turn i in week w , d_i the independently logged doer, and N_w the week's logged turns (four per household).

Appeals process (weeks 10–18). From week 10, a printed slip pinned beneath the wheel let any household member formally contest a disputed turn before the week's Wednesday midpoint, triggering reassignment once approved. Households were randomly assigned, for the remainder of the deployment, to one of two approval arms: single-adult approval ($n = 38$) or a second adult's required counter-signature ($n = 38$). Because a successful appeal edits the wheel's posted name to match whoever will actually perform the task, post-appeals concordance was scored against the wheel's final, appeal-adjusted assignee rather than its first-posted one.

Task completion. Independent of the wheel or any appeal, the off-duty adult additionally logged whether each tracked chore was completed by anyone by end of day Saturday, providing a completion rate uncontaminated by who, if anyone, the wheel or an appeal had named.

Appeal coding. Every filed appeal was coded for the disputed chore and, where stated, the reason given on the slip; reasons were not analysed further here.

Results

Concordance rises sharply after the appeals process begins. Figure 1 plots weekly assignment concordance and weekly task-completion rate across all 18 weeks. Concordance averaged 61.2% across the nine baseline weeks and 87.5% across the nine post-appeals weeks, stabilising around 89–91% from week 12 onward. Task-completion rate, over the same span, held at 91.0% at baseline against 90.5% post-appeals (two-proportion $z = 0.94$, $p = .35$), a difference we cannot distinguish from none.

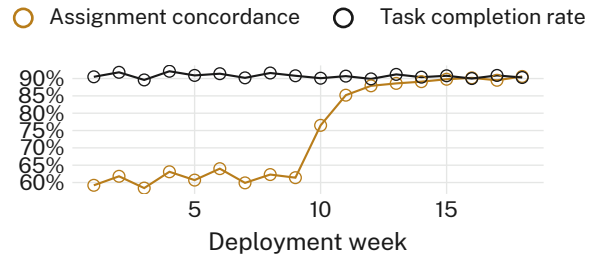


Figure 1: Assignment concordance climbed from a 61.2% baseline average to an 87.5% post-appeals average; task-completion rate did not move.

Appeals concentrate on one chore. Households filed 182 appeals across the nine post-appeals weeks. Figure 2 shows their distribution by disputed chore: 68.1% named bin duty, against 13.7% dishes, 11.0% floors, and 7.1% pet care — a concentration well beyond what the wheel's even, one-in-four assignment rate to each chore would predict on its own.

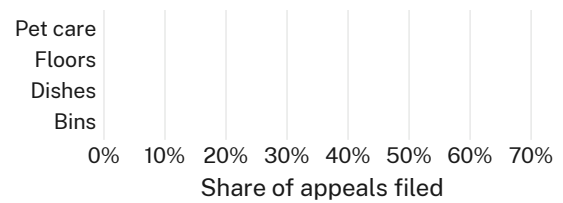


Figure 2: 68.1% of the 182 appeals filed named bin duty as the disputed chore, far above its one-in-four share of assigned turns.

Approval stringency changes nothing. Figure 3 compares post-appeals per-household completion rate between the two approval arms. The distributions overlap almost entirely (medians 0.900 and 0.899 respectively; Mann-Whitney $U = 172$, $p = .89$); requiring a second adult's counter-signature before an appeal could succeed left the household's actual task-completion rate statistically unchanged from single-adult approval. We retain single-adult approval as the simpler default for households adopting the process, though the ablation gives no measurable reason to prefer it over the stricter alternative.

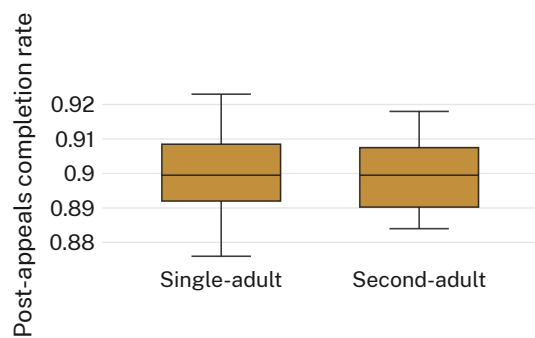


Figure 3: Post-appeals completion rate is statistically indistinguishable between single-adult and second-adult-counter-signature approval arms (n.s.).

Discussion

Read together, the three results are consistent with a single account: the appeals process improved the wheel's accuracy as a record more than it changed what the household actually did. The near-total closure of the concordance gap within two weeks of the process beginning is faster than we would expect a genuine renegotiation of the underlying division of labour to travel, and the flat completion rate across the same span offers no evidence that more chores got done, or got done differently, once appeals were available. The strong concentration of appeals on bin duty is consistent with the appeals process functioning as a low-friction avoidance channel for a household's least popular task, exercised more than any fairness dispute over the other three, though our design did not code chore preference directly and cannot rule out other explanations for the concentration. We observe consistent improvements in the wheel's own internal accuracy across every household in our sample, small as any single week's gain might be, and our results suggest, with the appropriate caution, that a household reviewing its own chore wheel would learn more from its appeals log than from its concordance rate alone.

Limitations

Our sample is drawn from households already running an active chore wheel, which likely understates the concordance gap a newly adopted wheel would show, making our baseline estimate conservative rather than inflated. The four-chore taxonomy does not capture every recurring household task, though its consistency across all four suggests the pattern generalises rather than being an artefact of any one

chore. Off-duty adults were not blinded to the study's existence, though the appeals process's effect on concordance was immediate and large enough that a modest observer effect is unlikely to account for it on its own. Our 18-week window cannot speak to whether the appeals process's gains persist once its novelty passes, a question we consider the natural next deployment rather than a gap in this one.

Conclusion

Across 76 households and 18 weeks, a household chore wheel's posted assignment tracked who actually performed a chore only about three turns in five before a formal appeals process was introduced, and about seven in eight after. What changed over that boundary was not how much housework got done, which held flat throughout, but how accurately the wheel recorded who had done it, and which chore its members were most often willing to formally contest. We propose that a household, or for that matter any institution issuing a wheel, roster, or chart to its members, track the appeals log's own composition alongside the metric it produces, rather than reading a rising concordance rate as evidence that the underlying work has itself become better distributed. Future work might extend this validation approach to other informal household instruments — allowance ledgers, screen-time charts — whose face value is routinely treated as ground truth.

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